

Faecal microbial flora and disease activity in rheumatoid arthritis during a vegan diet.

To clarify the role of the faecal flora in the diet-induced decrease of rheumatoid arthritis (RA) activity, 43 RA patients were randomized into two groups: the test group to receive living food, a form of uncooked vegan diet rich in lactobacilli, and the control group to continue their ordinary omnivorous diets. Based on clinical assessments before, during and after the intervention period, a disease improvement index was constructed for each patient. According to the index, patients were assigned either to a group with a high improvement index (HI) or to a group with a low improvement index (LO). Stool samples collected from each patient before the intervention and at 1 month were analysed by direct stool sample gas-liquid chromatography of bacterial cellular fatty acids. This method has proved to be a simple and sensitive way to detect changes and differences in the faecal microbial flora between individual stool samples or groups of them. A significant, diet-induced change in the faecal flora ($P = 0.001$) was observed in the test group, but not in the control group. Further, in the test group, a significant ($P = 0.001$) difference was detected between the HI and LO categories at 1 month, but not in the pre-test samples. We conclude that a vegan diet changes the faecal microbial flora in RA patients, and changes in the faecal flora are associated with improvement in RA activity.